Prabhjot Bhandal

905-598-9404 | bhandp9@mcmaster.ca | Hamilton, ON | in/prabhjot-bhandal | Portfolio

Technical Skills

Programming: C/C++, Python, Java, JavaScript, HTML/CSS, YAML
Hardware & Design: Altium Designer, KiCad, Arduino, ESP32, Soldering, Autodesk Inventor, FDM 3D Printing
Tools: Git, GitHub, Azure DevOps, Docker, Portainer, Microsoft Copilot, Microsoft Loop

Education

McMaster University Hamilton, ON
Bachelor of Engineering (B.Eng.), Mechatronics Engineering & Society *Expected 2029*

Experience

Avionics Subteam Member June 2026 – Present
McMaster Aerospace Team *Hamilton, ON*

- **Reviewed part dimensions and interfaces** across 4 subsystems, raising concerns and proposing mitigations
- **Proposed a battery heat-exchange system** and investigated heat sink materials and mechanical design; dropped it when it violated competition rules
- **Shifted to battery selection**, planning **internal-resistance and discharge-rate tests** to compare candidates

Student Project Coordinator, Group Technology Modernization May 2026 – Aug. 2026
Equitable *Waterloo, ON*

- **Coordinated 4 sprints for a 30+ person team**, reporting productivity for each sprint and maintaining the **Azure DevOps backlog** with custom tasks, documented procedures, and changeover information
- **Acted as project manager** for **9 days**, leading all meetings, assigning work with technical leads, supporting UI testing and fixes and training the incoming project coordinator
- **Built Copilot agents** and a **Loop** board for meeting notes and action tracking, logging 600+ leadership items

Electrical Team Lead Feb. – June 2026
McMaster Aerial Robotics & Drone Team *Hamilton, ON*

- **Co-planned a 4-month electrical build** for 8 members; **overhauled task distribution** into small, modular tasks
- **Designed and simulated a buck converter in Altium Designer** to step a 12S battery supply down to **5 V logical power**
- **Flagged high current draw** in the drone's power delivery; recommended thicker wire and shorter connections

Projects

Synabond – Habit Tracker Virtual Pet | *C/C++, ESP32, I^2C* Winter 2026

- Built an **ESP32** habit tracker with an SSD1306 OLED, DS3231 RTC (shared **I^2C** bus), buttons, and RGB LED
- Implemented **2 state machines** with interrupt-driven buttons and a **Hebbian weight matrix** (+0.05 co-completion, 0.99 daily decay) to find keystone habits and schedule reminders
- Built a five-state **pet mood system** and ran **10+ subsystem and edge-case tests** to debug timing and hardware faults

The Skovel – Modular Snow Travel Multitool | *Autodesk Inventor, FDM 3D Printing* Winter 2026

- Worked in a team to **model, 3D-print and assemble** a foldable ski-shovel concept using **Autodesk Inventor**
- **Presented the problem, design rationale and constraints** in plain language to a non-techinal audience
- **Delegated tasks by team strengths** in a 24-hour designathon and helped subteams resolve design questions

Haptic Smart Cane for Crosswalk Guidance | *C/C++, Arduino, IR Sensor, ADXL345* Feb. – Apr. 2025

- Contributed to a **team-built Arduino attachment** guiding a deafblind user across crosswalks with haptic feedback
- Integrated an **IR sensor** to distinguish crosswalk and road surfaces and an **ADXL345 accelerometer** to detect tilt-based veering
- Developed **C/C++ calibration and angle-processing logic** that triggers direction-coded vibration, contributing to a reported **40% improvement in crosswalk alignment accuracy**